Payload-Aware Two-Tier UAV Route Planning for Last-Mile Delivery

Review Report – II

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First Review – Queries and Answers

The panel raised one concern with two parts at Review I:

Query 1

The overall technical complexity of the proposed system was too low for a final-year project.

Query 2

Given that complexity, a Phase I that runs until December could not be justified.

Both are fair readings of the Review I prototype. Since then, the airspace graph and cost model — the foundation every later stage depends on — have been rebuilt.

Query 1 → A richer 3-D airspace lattice and a directional wind-risk model (next slide)

Query 2 → Two coupled, repeated loops — benchmarking/evaluation and simulation fidelity — remain (slide after)

Query 1 — A Harder Airspace Graph & Wind Model

Two changes account for most of the added complexity since Review I:

1. Multi-source node selection

- 3-D lattice (200 m spacing, 6 km box) replaces OSM road-only graph — 900 candidate nodes
- Elevation from a 4-provider DEM chain; building heights from OSM footprints
- Legal ceiling check: $\max(60\,\text{m},\ h_{\text{bldg}} + 15\,\text{m}) \leq 120\,\text{m}$ (Drone Rules, 2021)
- Restricted-zone polygons remove nodes and clipping edges

2. Directional wind-aware risk

- Wind split per edge into headwind + crosswind components
- 11-component weighted risk score (wind, gust, turbulence, rain, visibility, population, buildings. . .)
- Weather gate holds all flights outside safe limits
- Edge costs become directional: $u \to v \neq v \to u$ under wind

+27.8%
energy for east-bound vs west-bound edges, 25 km/h east wind

6×
higher crosswind risk on N–S edges vs E–W edges (0.225 vs 0.038)

36 → 6–15
routes on the Pareto front as calm air turns into a steady wind

Module 1 in Detail: Node Selection & Wind Decomposition

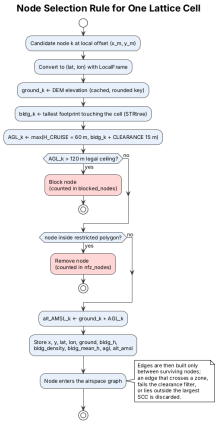


Fig. 4.5 – Selection rule applied to each candidate lattice node

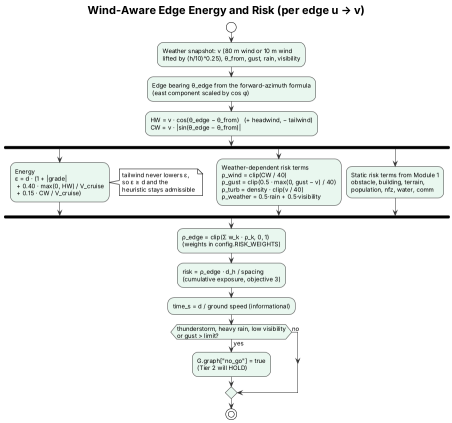


Fig. 4.6 – Per-edge wind decomposition into energy and risk

Query 2 — Why Phase I Runs Through December

Two pieces of work remain, and neither is a one-off task — both are repeated loops.

Loop A — Benchmark, Evaluate, Improve

- 36 cases: 4 graphs × 9 wind directions
- NAMOA*, 3 single-objective A*, weighted-sum A*, and MOPSO (100 particles × 200 iters × 30 seeded reps) per case — ∼21.6M particle updates/pass
- 225 payload-tag combinations × 4 selection rules per case
- First pass already surfaced 5 issues needing a fix + full re-run (density-label bug, monotonicity bug, pooled MOPSO stats, front-collapse question, placeholder weights)
- HV gap vs MOPSO on 200-node graph is only 0.0004 — needs 30 reps + a statistical test before any claim is safe

Loop B — Simulation Fidelity

- SITL mission chain already flies correctly (cross-track RMS 0.11–0.13 m) — but only with ArduPilot's built-in physics, not the city model
- Gazebo world must be built from live OSM + DEM and pass load/rest/real-time gates
- Gazebo has no wind yet, while the planner already plans for wind
- Battery model constants are placeholders — must be calibrated from simulated flights per payload class
- Every mismatch between planned and flown route feeds back into Loop A — the two loops are coupled and must run side by side

The Benchmark – Evaluate – Improve Cycle (Oct–Dec)

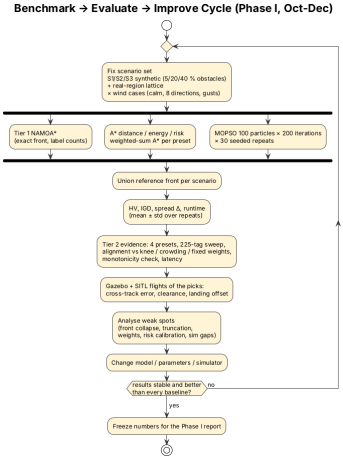


Fig. 2.1 – The benchmark, evaluate and improve cycle planned for October to December

Problem Statement

The Problem

Existing UAV planners apply a fixed geometric Pareto-selection rule, or a hand-tuned weighted-sum scalar, independent of payload type. When the payload changes — fragile sample vs. emergency medicine vs. bulk relief — an expert must re-tune weights and re-run the search from scratch. That intervention is infeasible in the field.

Formally

- Given: an airspace graph G with edge costs $c(e) = [\text{distance}, \text{energy}, \text{risk}]$; a pre-computed Pareto front P over G; and a payload tuple $(F, W, U, P_{\text{lvl}})$ — fragility, weight class, urgency, priority
- Select the route $p^* \in P$ that best satisfies the payload's implied routing requirements
- Without expert re-tuning, and without re-running the multi-objective search

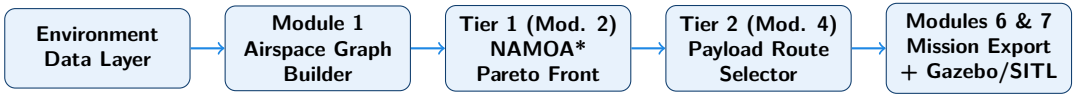
Constraints & success criteria

- Selection must be deterministic, reproducible, and complete within 1 second of tag entry
- The selected route's distance/energy/risk must match what the payload tag implies — checked by a Tier 2 alignment metric

Project Objectives

1. Formulate multi-objective UAV route planning as a Pareto front pre-computation task, distinguishing deterministic graph-search from stochastic population-based alternatives
2. Design a payload metadata schema and rule-based mapping from mission context to Pareto-point selection weights, with no expert tuning
3. Benchmark Tier 1 route quality and front coverage against single-objective A* and MOPSO using hypervolume, spread and IGD
4. Quantify Tier 2 selection accuracy via payload-alignment across mission types (fragile sample, emergency medicine, disaster relief)
5. Establish novelty: payload metadata as the sole, deterministic input to Pareto-point selection, replacing context-blind entropy/diversity criteria

Proposed System Architecture



Module 3 — Baseline Planners

Single-objective A* $\times$ 3, weighted-sum A*, MOPSO — run beside the main path for comparison

Module 5 — Evaluation & Benchmarking

Hypervolume, IGD, spread, Tier 2 alignment, 225-tag sweep, ablation — scores every stage

Key idea: the Pareto front is built once per region and weather window (Tier 1) and reused across missions; a new mission runs only Tier 2, in well under a second.

Hard constraints enforced throughout

Cruise $\geq$ 60 m above ground & 15 m above roofs • $\leq$ 120 m AGL (Drone Rules, 2021) • no entry into restricted zones • route energy within 80% of battery capacity

System Architecture (Fig. 4.2)

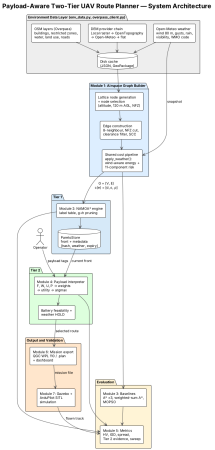


Fig. 4.2 – System architecture: Tier 1 builds and stores the front per region and weather window; Tier 2 selects a route per payload tag

System Modules

1. **Module 1 – Airspace Graph Builder:** 3-D lattice from DEM, OSM buildings, restricted zones; shared [distance, energy, risk] cost pipeline
2. **Module 2 – Tier 1 — NAMOA* Engine:** Exact multi-objective A* search with a label table and admissible vector heuristic
3. **Module 3 – Baseline Planners:** Single-objective A* $\times 3$, weighted-sum A*, and a MOPSO baseline
4. **Module 4 – Tier 2 — Payload Selector:** Maps 4 payload tags $\rightarrow$ weights $\rightarrow$ one route; battery feasibility + weather HOLD
5. **Module 5 – Evaluation & Benchmarking:** HV, IGD, spread, Tier 2 alignment, 225-tag sweep, ablation studies
6. **Module 6 – Mission Export & Dashboard:** QGC WPL 110 / .plan export; Tkinter operator dashboard
7. **Module 7 – Simulation & SITL Validation:** Gazebo Harmonic + ArduPilot SITL; cross-track error, clearance, landing offset

Methodology / Workflow

1. Collect & cache environment data (weather, terrain, OSM) for the delivery region
2. Build the airspace graph: node & edge selection, shared cost pipeline
3. Tier 1 computes the Pareto front; store it with graph hash + weather snapshot
4. Tier 2 reads a payload tag and returns one route from the current front
5. Export the route as a mission; fly it in Gazebo + ArduPilot SITL
6. Baselines & evaluation score every stage; findings feed back into the model

Methodology Overview & Mission Sequence

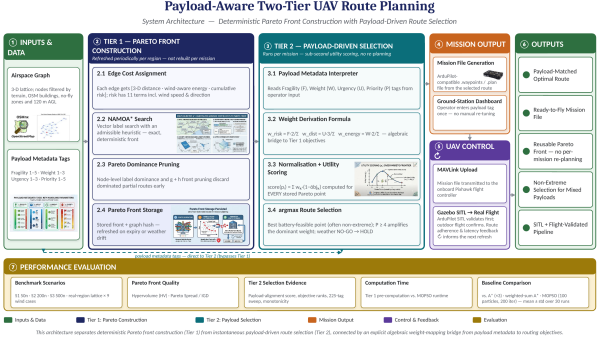


Fig. 4.1 – Methodology overview: Tier 1 front construction feeding Tier 2 route selection

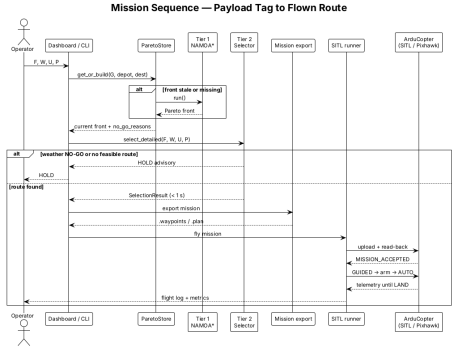


Fig. 4.3 – Mission sequence from payload tag entry to the flown route

Core Algorithms — NAMOA* & Payload Selection

Tier 1 — NAMOA* (exact Pareto search)

- One label (node, g, parent) per non-dominated partial path, held in a flat table — no path copying
- Admissible vector heuristic $h(v) = [h, h, 0]$, h = great-circle distance to target
- A label is pruned when the current front weakly dominates $g + h$
- Reports a truncation flag + expansion/generation diagnostics if a search cap is hit
- Guarantees: IGD = 0 on every tested scenario — the front contains every reference point

Tier 2 — Payload-Driven Selection

- $r_{\text{raw}} = F \times 2$, $d_{\text{raw}} = U \times 3$, $e_{\text{raw}} = W \times 2 \rightarrow$ normalised into $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$
- High priority ($P_{\text{lvl}} \geq 4$): dominant weight amplified by $0.1 \times w_{\max}$, then renormalised
- Min-max normalise every route's (d, ε, ρ) across the front
- $\text{score}(p) = w_{\text{dist}}(1 - \tilde{d}) + w_{\text{energy}}(1 - \tilde{\varepsilon}) + w_{\text{risk}}(1 - \tilde{\rho})$; $p^* = \arg\max \text{score}$
- Battery-infeasible routes removed first; weather HOLD returned if no route survives

< 1 ms	IGD = 0	0 / 225
Tier 2 selection latency (target was < 1 s)	NAMOA* front vs. reference, every scenario	monotonicity violations across the full tag sweep

Tier 1 & Tier 2 Activity Flows

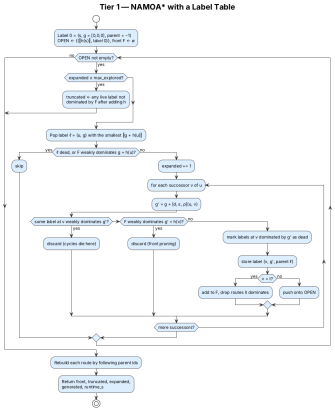


Fig. 4.7 – Tier 1: NAMOA* with a label table

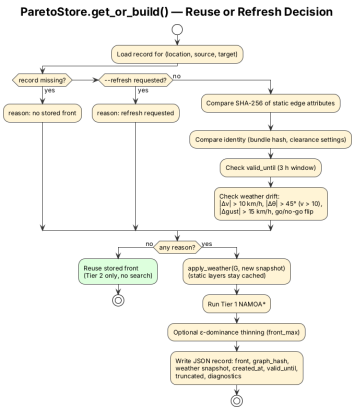


Fig. 4.8 – Front store: reuse or rebuild

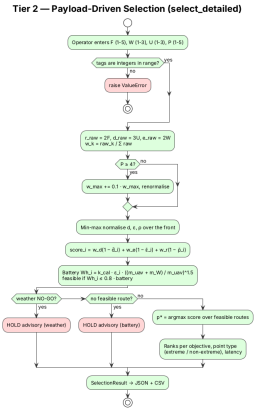


Fig. 4.9 – Tier 2: validation, feasibility, weather hold

Payload Metadata → Routing Weights

Each of the four payload tags maps to exactly one Tier 1 objective — a deterministic algebraic bridge, not an expert-set rule:

Payload tag	Maps to	Reasoning
Fragility (F), 1–5	w_{risk}	Fragile items need low-vibration, low-risk corridors
Payload weight (W), 1–3	w_{energy}	Heavier payload → higher hover/lift power draw
Delivery urgency (U), 1–3	w_{dist}	Time-critical deliveries need the shortest path
Priority level (P_{lvl}), 1–5	amplifier	Scales the dominant weight when scores are close

Reference presets & derived weights

Preset	w_{risk}	w_{dist}	w_{energy}
Fragile medical	0.667	0.200	0.133
Emergency medicine	0.144	0.712	0.144
Disaster relief	0.182	0.273	0.545
Mixed cargo	0.375	0.375	0.250

Only emergency medicine has $P_{\text{lvl}} \geq 4$, so it is the only preset whose dominant weight (distance) is amplified.

Implementation Status

Module	Status	Evidence
1 — Airspace Graph Builder	Complete	Lattice + cost pipeline running on real & synthetic graphs
2 — Tier 1 NAMOA*	Complete	Front equals brute-force front on small graphs; 0 dominated/duplicate routes
3 — Baseline Planners	Complete	3× A*, weighted-sum A*, MOPSO implemented and run
4 — Tier 2 Selector	Complete	Weight bridge, feasibility filter, weather HOLD, JSON/CSV record
5 — Evaluation Suite	Complete (1st pass)	HV/IGD/spread, 225-tag sweep, ablation — report-scale run in progress
6 — Mission Export & Dashboard	Complete (2 open defects)	QGC WPL 110 export verified by round-trip test; Tkinter dashboard functional
7 — Simulation & SITL	Core flown, fidelity pending	SITL mission chain verified; Gazebo wind & battery calibration next

∼265
automated tests (planner + simulation suites)

0.11–0.13 m
cross-track RMS error, flown vs. planned route in SITL

Oct–Dec
remaining benchmarking + simulation-fidelity cycle

Simulation Pipeline & Core Classes

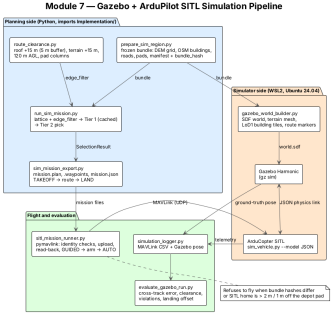


Fig. 4.10 – Gazebo and ArduPilot SITL simulation pipeline

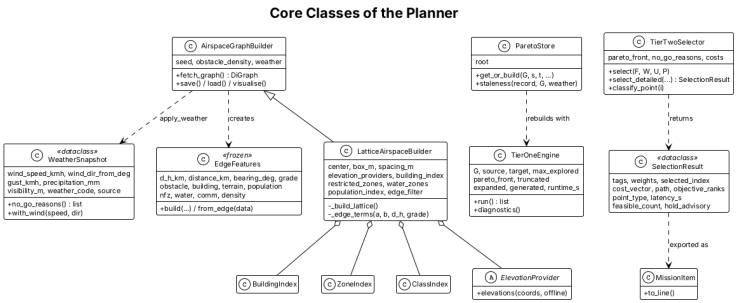
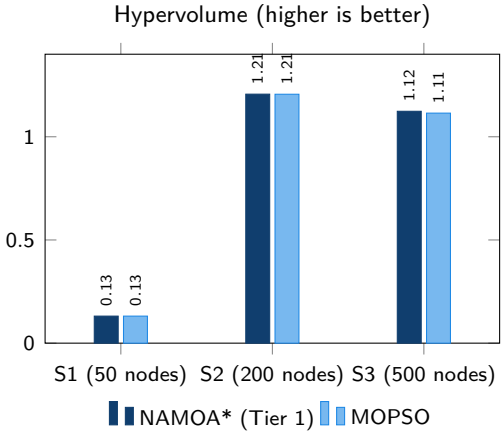


Fig. 5.1 – Core classes of the planner

Results — Tier 1 Front Quality vs. Baselines

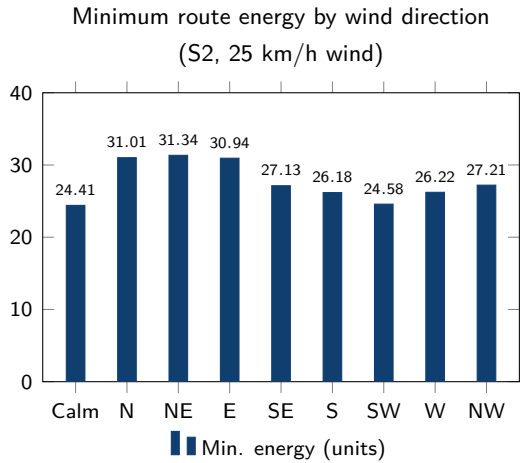


What the exact IGD = 0 result means

- NAMOA* front contains every point of the reference front on every scenario (IGD = 0)
- MOPSO recovers only 12 of 16 routes on S2, and 38 of 115 on S3
- HV gap stays small (0.0004 on S2, 0.009 on S3) because MOPSO still spans the extremes
- Missed interior points are exactly what Tier 2 needs for mixed payloads
- Cost of exactness: 33.8 s (NAMOA*) vs. 4.3 s (MOPSO) on S3 — acceptable since Tier 1 runs once per region/weather window, not per mission

Bottom line: the exact method wins on front coverage and spread, where it matters for payload-aware selection — not just on the headline HV number.

Results — Effect of Wind Direction on the Front



Why direction — not just speed — matters

- Same wind speed: cheapest route needs 24.58 energy units (SW wind) vs. 31.34 (NE wind) — a 27.5% swing
- Opposite directions give the same minimum risk (crosswind depends only on the wind's axis) but very different minimum energy (headwind depends on direction along that axis)
- East-bound edges cost 27.8% more energy/km than west-bound edges under an east wind
- N–S edges carry 6× the crosswind risk of E–W edges (0.225 vs. 0.038)

Open question for the next pass: under steady wind, three of four payload presets converge on the same route on the benchmark graph — is that the right answer, or an artefact of min-max normalisation on a short front?

Contribution of Each Risk Layer (Ablation)

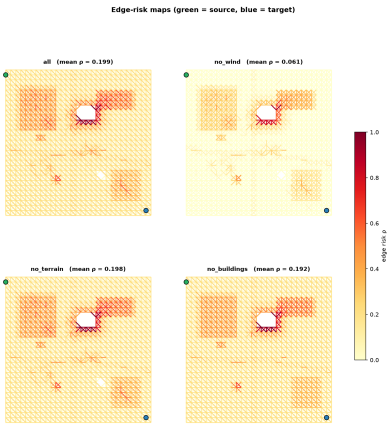


Fig. 6.4 – Edge risk maps for the ablation variants (fixture layers)

Table 6.5 – 30 km/h wind from SW, 6 km lattice

- All layers: 196 routes, mean edge risk 0.199
- No wind: 205 routes, mean edge risk 0.061 (−69%); safest route risk drops from 8.85 to 0.10
- No terrain: only 56 routes — altitude changes are what separate otherwise similar paths
- No buildings: 190 routes, higher max risk (13.08 → 12.26 removed)
- Wind is the single largest driver of risk; terrain mostly drives front size; buildings raise the upper end of the risk range

Results — Tier 2 Selection & Simulation

Preset (S3, 115-route front)	Rank: dist	Rank: energy	Rank: risk	Point type
Fragile medical	91	81	25	non-extreme
Emergency medicine	22	26	94	non-extreme
Disaster relief	28	1	88	min-energy
Mixed cargo	84	59	32	non-extreme

All four presets pick four different, mostly non-extreme routes on the large front — trade-offs a single-objective or fixed-weight planner cannot make.

1.9%
longer than the shortest route — emergency medicine's pick, for a much lower risk

$<$1 ms
every Tier 2 selection, far inside the 1 s target

Simulation (Gazebo + ArduPilot SITL)

- Mission upload, read-back, switch to AUTO, landing and disarm all succeeded
- Cross-track RMS error: 0.11–0.13 m
- Landing offset from destination pad: 0.01–0.02 m
- Confirms the mission chain from Tier 2 to the autopilot is correct, including the altitude frame
- Not yet covered: the 3-D city model, wind in the simulator, and calibrated battery constants

Tier 2 Routes on the Lattice & SITL Simulation (Fig. 6.5)

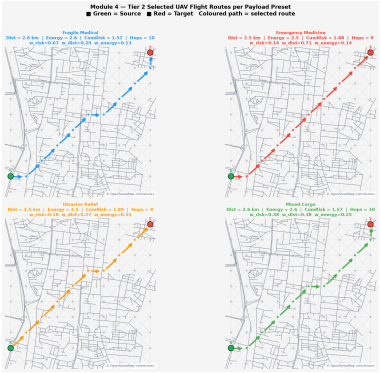
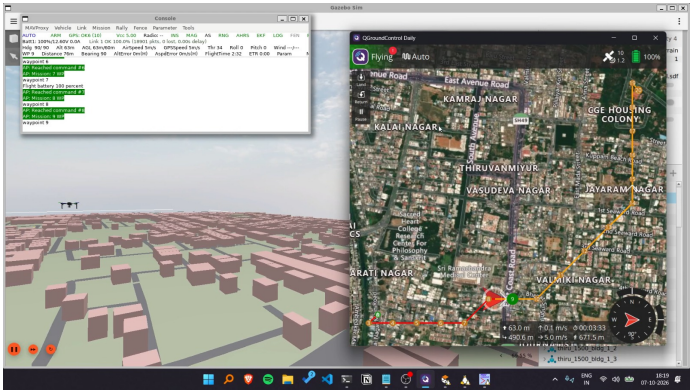


Fig. 6.5 – Tier 2 routes for the four presets on a 2 km lattice



Gazebo Harmonic + ArduPilot SITL flying an uploaded Tier 2 mission in AUTO — MAVProxy console (left), QGroundControl mission replay (right)

Comparison with Existing Work

Property	Chokkalingam & Ganesan	Tsoutsouras et al.	Ren et al.	This Work
Front construction	POPSO (stochastic)	None	CDNSGA-II (stochastic)	**NAMOA* — exact, deterministic**
Final selection	Weighted sum, re-tuned per mission	Not applicable	Crowding distance	**Utility from payload tags, argmax over full front**
Payload in selection	No	Mass in energy model only	No	**Fragility, weight, urgency, priority**
Wind direction in cost	Not reported	Not reported	Not reported	**Headwind + crosswind per edge, gust, weather gate**
Front reuse	Re-run per mission	Not applicable	Re-run per mission	**Stored front, drift-based refresh**

The comparison is qualitative for the cited works (code/data not public); their selection rules (crowding distance, fixed weights) are applied to our fronts for the quantitative comparison, reported above with the payload-alignment metric.

Discussion & Next Steps

Supported so far

- The exact Tier 1 front covers the reference front completely, while MOPSO recovers only a third of it on the largest scenario
- Tier 2 picks non-extreme routes for mixed payloads in well under a millisecond, and different payloads get different routes on a large front
- The directional wind model changes energy, risk and front shape as physics predicts — a direction-blind model would miss a 27.5% energy difference at a fixed wind speed

Next in the benchmark + simulation cycle

- Run MOPSO at report-scale settings with per-repetition statistics
- Study the front collapse under steady wind (normalisation vs. weight multipliers)
- Calibrate risk weights and battery constants from simulated flights
- Fix the density-label defect and re-group the monotonicity check by scenario
- Repeat ablation & Tier 2 evaluation on live OpenStreetMap data
- Add wind to the Gazebo world; fly each preset under multiple wind cases

Team Responsibilities

Team Member	Responsibilities	Current Status
Thirumurugan RA	Software simulation: Gazebo Harmonic world builder from OSM + DEM, ArduPilot SITL mission runner, QGC WPL 110 / .plan mission export, cross-track and landing-offset measurement, Gazebo wind and battery calibration	SITL chain verified; simulation fidelity in progress
Meghna Manimaran	3-D airspace lattice with multi-source node selection (DEM chain, OSM buildings, restricted zones, cached Overpass client), directional wind-aware energy and 11-component risk model, weather go/no-go gate, Tier 1 rewrite as NAMOA* with label table	Complete
Shrika SP Thota	Front store with graph hash and weather-drift refresh, Tier 2 validation, battery feasibility and weather HOLD, baseline planners (A* $\times$3, weighted-sum A*, MOPSO), evaluation suite (HV, IGD, spread, 225-tag sweep, ablation), automated tests	First pass complete; report-scale run in progress

Phase I (through December) is a shared, coupled effort: the benchmarking/evaluation loop and the simulation-fidelity loop (Chapter 2) draw on all three members across weight calibration, baseline comparison and SITL validation.

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Thank You